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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

# Transmission of Signal Through Discrete Time Lti Systems

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

| 6 | 6 | 1988 – 2024 | English |
|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | LANGUAGE |

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## Marks 1 2 questions

**Q1.** GATE ECE 2024

A causal and stable LTI system with impulse response *h(t)* produces an output *y(t)* for an input signal *x(t)*. A signal *x(0.5t)* is applied to another causal and stable LTI system with impulse response *h(0.5t)*. The resulting output is ____.

Open solution

(A) 2*y(0.5t)* (B) 4*y(0.5t)* (C) 0.25*y(2t)* (D) 0.25*y(0.25t)*

**Q2.** GATE ECE 2015 Set 3

Consider a four-point moving average filter defined by the equation $y[n] = \sum_{i=0}^{3} \alpha_i x[n-i]$. The condition on the filter coefficients that results in a null at zero frequency is

Open solution

(A) $\alpha_1 = \alpha_2 = 0;\ \alpha_0 = -\alpha_3$ (B) $\alpha_1 = \alpha_2 = 1;\ \alpha_0 = -\alpha_3$

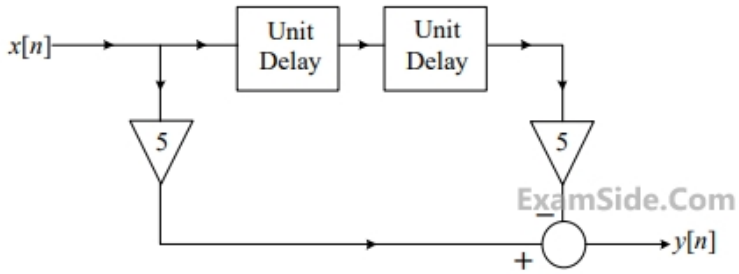
(C) $\alpha_0 = \alpha_3 = 0;\ \alpha_1 = \alpha_2$ (D) $\alpha_1 = \alpha_2 = 0;\ \alpha_0 = \alpha_3$

## Marks 2 3 questions

**Q3.** GATE ECE 2016 Set 3

The direct form structure of an FIR (finite impulse response) filter is shown in the figure.

Open solution




The filter can be used to approximate a

(A) low-pass filter (B) high-pass filter (C) band-pass filter (D) band-stop filter

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**Q4.** GATE ECE 2009

A system with transfer function H(z) has impulse response h(.) defined as h(2) = 1, h(3) = - 1 and h (k) = 0 otherwise. Consider the following statements.
S1: H(z) is a low-pass filter.
S2: H(z) is an FIR filter.
Which of the following is correct?

(A) Only S2 is true

(B) Both S1 and S2 are false.

(C) Both S1 and S2 are true, and S2 is a reason for S1.

(D) Both S1 and S2 are true, but S2 is not a reason for S1.

Open solution

**Q5.** GATE ECE 2005

A signal x(n)$= \sin(\omega_0 n + \phi)$ is the input to a linear time-invariant system having a frequency response $H(e^{j\omega})$.If the output of the system is $Ax(n - n_0)$, then the most general form of $\angle H(e^{j\omega})$ will be

(A) $-n_0\omega_0 + \beta$ for any arbitrary real $\beta$

(B) $-n_0\omega_0 + 2\pi k$ for any arbitrary integer k

(C) $n_0\omega_0 + 2\pi k$ for any arbitrary integer k

(D) $-n_0\omega_0 + \phi$

Open solution

## Marks 4 1 questions

**Q6.** GATE ECE 1988

A signal x(t) = $\exp(-2\pi Bt)\, u(t)$ is the input to an ideal low pass filter with bandwidth B Hz. The output is denoted by y(t). Evaluate $\int_{-\infty}^{\infty} [y(t) - x(t)]^2\, dt$.

Open solution

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# Answer Key

## Marks 1

| | |
|---|---|
| 1. **A** | 2. **A** |

## Marks 2

| | | |
|---|---|---|
| 3. **C** | 4. **A** | 5. **B** |

## Marks 4

| |
|---|
| 6. **solve it.** |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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